

ZEXUAN WANG

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Research interests: World models, embodied AI, latent-action learning, and human motion / video generation.

Hometown: Xi'an, Shaanxi, China | Age: 21



EDUCATION

Chang'an University

Sep 2023 - Jun 2027 (expected)

B.Eng. candidate in Intelligent Manufacturing Engineering | GPA: **4.14/5.0** | Major rank: **1/122**.

Beijing Institute of Technology & Qiyuan Laboratory

2027 - 2032 (expected)

Incoming Ph.D. student in **Computer Science and Technology**, starting in 2027.

Research advisor: Junliang Xing, Tsinghua University.

RESEARCH EXPERIENCE

Tsinghua University, Qiyuan Laboratory

Sep 2026 - Present

Research Assistant | Advisor: Junliang Xing

- Conducting **early-stage literature review** on world models for multi-agent interaction and decision-making; exploring links between generative dynamics models, latent actions, and embodied agents.

Tsinghua University, Institute for AI Industry Research (AIR)

May 2026 - Present

Research Intern | Advisor: Hao Zhao

- Working on real-time digital-human generation and inference acceleration, including LongCat-Video and LongCat-Video-Avatar-1.5; studying text/audio conditioning, DiT, Flow Matching inference, and LoRA loading.
- Implementing and debugging **multi-audio Avatar inference**, KV-cache reuse, and sparse-attention acceleration modules; reproducing the MolmoMotion data-annotation pipeline for a physical-AI project.

Tsinghua Shenzhen International Graduate School

Oct 2025 - Jun 2026

Research Intern | Xiu Li's Group | Mentor: Ronghui Li

- Researched text- and sparse-keyframe-conditioned motion generation using DDPM, Flow Matching, and SMPL-X; led **PAIC and MARC-Flow**, covering model implementation, baseline reproduction, evaluation, ablations, and writing.
- Contributed motion-data conversion, garment simulation, and 3D rendering to a music-conditioned duet-dance project using Blender, CLO, and Unreal Engine.

Chang'an University

Feb 2025 - Sep 2025

Undergraduate Research Assistant | Advisor: Xiaohua Xia

- Independently carried out a first-author study on content-robust no-reference defocus estimation, from problem formulation and synthetic-data generation to model design, cross-dataset experiments, and manuscript preparation.
- Implemented the model backbone, data augmentation, and hybrid evolutionary hyperparameter search; open-sourced **EAKCNet-F**. Project support: NSFC 61901056; Shaanxi 2025CY-YBXM-545; Shaanxi Transport 24-74K.

TECHNICAL SKILLS

Programming: Python, PyTorch, MATLAB; familiar with C++. **Development:** Git, Linux.

Modeling: Diffusion models, Flow Matching, conditional generation, SMPL-X, KV caching, sparse attention.

3D / engineering tools: Blender, CLO, Unreal Engine, Visio, SolidWorks, ANSYS, ideaMaker.

English: CET-6: **523**; CET-4: **532**.

RESEARCH PROJECTS & MANUSCRIPTS

Content-robust no-reference defocus estimation with asymmetric-attention learning and sigma calibration

Zexuan Wang, Xiaohua Xia | First-author manuscript | Under review at Pattern Recognition

- **Problem / method:** Addressed content-dependent focus scores and discrete blur-level labels. Built multi-level Gaussian-defocus data from Waterloo images and continuous sigma-based labels; designed asymmetric dual-branch attention to isolate edge-spread and texture-attenuation cues from content bias, with cross-scale fusion and score calibration.
- **Contribution / results:** Led the full research and writing process, including robustness and interpretability analyses. Improved score correlations on TID2013 and KADID-10k and cross-dataset generalization by separating defocus cues from image content.

Code: github.com/glkssK/EAKCNet-F

RESEARCH PROJECTS & MANUSCRIPTS (CONTINUED)

Music Conditioned Duet Dance Generation and Dance Accompaniment

Collaborative research project | Motion-data processing and 3D visualization | Dec 2025 - Mar 2026

- **Problem / approach:** Addressed limited long-duration duet-dance data and inconsistencies in motion interaction, clothing simulation, and rendering. The project paired music with SMPL-X duet motion and used contact labels and inter-person geometry to improve interaction modeling.
- **Contribution:** Converted PKL/NPZ and SMPL-X motion data and checked skeleton consistency. Built a rendering workflow covering Blender motion import and Alembic export, CLO garment binding and simulation, and Unreal Engine scenes, materials, lighting, cameras, and video rendering.
- **Results:** Contributed to a dataset of approximately **5.85 hours across 21 dance styles** and an end-to-end motion-to-video visualization workflow, improving format interoperability and the consistency of clothing simulation and presentation videos.

PAIC: Position-Aligned Interval Conditioning for Text-Guided Human Motion In-Betweening

Zexuan Wang, Zeyu Ling, Xitian Huang, Yujie Hou, Yiwei Sheng

First-author manuscript | Submitted to AAAI 2027 | Feb 2026 - Jun 2026

- **Problem:** Global keyframe/mask conditioning does not explicitly describe each missing frame's neighboring observations, boundary distances, or transition phase, making long-gap completion and text control difficult under changing keyframe layouts.
- **Method:** Explicitly encoded left/right boundary distances, relative phase, interval length, and support validity. Combined a refinable coarse transition proposal with separate anchor and text memories in a single-stage masked diffusion model.
- **Contribution / results:** Led formulation, implementation, training, and writing; established unified evaluation and independently ran comparisons, ablations, gap-length robustness tests, and fixed-keyframe semantic-response tests. On HumanML3D, achieved lower FID and boundary error than task-matched base models across observation densities.

MARC-Flow: Mask-Adaptive Residual Coordinates for Text-Conditioned Motion Completion

Zexuan Wang, Zeyu Ling, Xitian Huang, Yujie Hou, Yiwei Sheng

First-author manuscript | Submitted to AAAI 2027 | Mar 2026 - Jul 2026

- **Problem / method:** Separated keyframe-predictable structure from text-dependent residual motion. Used a frozen text-free prior, a lightweight mask-adaptive scale adapter, and residual Flow Matching to normalize residual scales across time, body parts, and keyframe layouts, then reconstructed motion with exact keyframe recovery.
- **Contribution:** Contributed to problem formulation and method design; implemented training and evaluation, reproduced strong baselines, and established fixed-K (2, 5, 20) and mixed-K (2-20) protocols. Conducted ablations, scale analyses, metric audits, failure-case analysis, visualization, and manuscript writing.
- **Results:** A single model covered **K = 2-20** without sparsity-specific retraining, test-time candidate selection, or trajectory post-processing. On HumanML3D, reduced FID by **66.2%, 89.8%, and 69.7%** at K = 2, 5, and 20 relative to the best compared method in each setting, with comparable motion diversity.

Additional publication: First author of one published English-language, EI-indexed conference paper.

HONORS & AWARDS

National Scholarship; Feng Zhongxu Scholarship.

- **National First Prize**, Global Campus Artificial Intelligence Algorithm Elite Competition, Industry Problem Track, **2025**.
- **National Second Prize**, Mechanical Engineering Innovation and Creativity Competition, Intelligent Manufacturing / Intelligent Operation and Maintenance Track, **2025**.
- **National Third Prize**, Mechanical Engineering Innovation and Creativity Competition, Intelligent Manufacturing / Intelligent Production Control Track, **2025**.
- **National Third Prize**, 18th Advanced Robotics and Simulation Technology Competition, **2025**.
- **National Second Prize**, Mechanical Engineering Innovation and Creativity Competition, Intelligent Manufacturing / Creative Track, **2024**.
- **National Third Prize**, The World of English Cup National College English Writing Competition, Undergraduate Division, **2024**.